SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5102

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

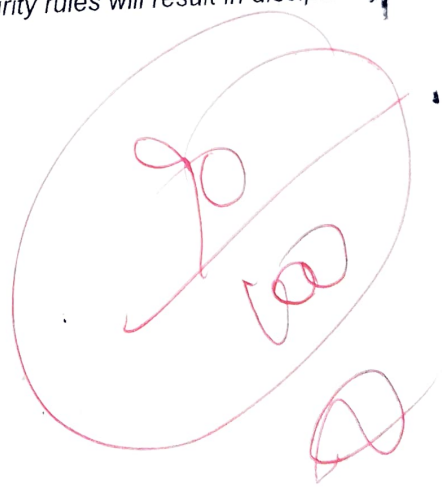
CONCEPT MAPPING

Code of Conduct:

I A. Midhuna (192521302) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

A. Midhuna

Signature of the student:



Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i. Map how each affects the Data structure and Security level
        ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
    a. Show how each concept is mathematically connected
    b. Map their role in encryption/decryption
    c. Analyze which concept is most critical for public key cryptography and justify

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Cryptography and Network Security

CO1 ASE2 - concept - (01)

1)

Security Attacks - Threats - Risks - Security Goals

Security Attacks
- * phishing
- * DDoS Attack
- * malware
- * SQL injection
- * man in the middle (MITM)

leads to

Threats
- * Credential theft
- * Service disruption
- * Data interception
- * Database manipulation

causes

Risk
- * unauthorized access
- * Downtime
- * Data loss
- * Sensitive data leakage
- * Information exposure

mitigated by

Security Goals (CIA)
(protect against threats and reduce risk)

Confidentiality
(prevents unauthorized access to data)
- * Encryption
- * Access control
- * multi factor authentication

Integrity
(prevents unauthorized modification of data)
- * Digital signature
- * checksums
- * Regular backups

Availability
(keeps system & system accessible)
- * DDoS protection
- * Redundant system
- * Disaster recovery plan

a) Attack → Threat → Risk (with Real-world Eg)

Security Attack	Specific Threat	Risk	Real world Examples
*phishing	credential theft	unauthorized account access	Fake bank email steals customer password
* DDoS Attack	service disruption	website becomes unavailable	Attack makes online shopping website inaccessible
* malware	Data compromise	Data loss or system damage	Ransomware encrypts hospital files
* man in the middle	Data interception	confidental information exposed	Attacker capture login details Insecure network
* SQL injection	Database manipulation	sensitive data leakage	Attacker access customer records through vulnerable website.

Conclusion:

Security attack can create threads & risk. The CIA security goals confidentality, integrity, Availability protect system prevent unauthorized access maintain accurate data, ensure service remain available. The relationship attacks, threats, risk and sec

2)

comparison of substitution Techniques (caesar, monoalphabetic, Vigenere, Hill)

caesar cipher
shifts each letter by fixed number of position alphabet

Key usage
single integer (shift value) eg: 3

complexity
very low (simple to implument)

security Strength
* weak (easy to break using brute force)

monoalphabetic cipher
used a fixed substitution mapping for all letters (eg: A → D, B → W, ...)

Key usage
26 letters substitution key (one for each alphabet)

complexity
Low to moderate (requires key management)

security Strength
*moderate (vulnerable to frequency analysis)

Vigenere cipher
Uses a repeating keyword with multiple caesar shifts

Key usage
Keyword (can be a word or random string)

complexity
moderate (more complex than caesar and monoalphabetic

security strength
Stronger than caesar and monoalphabetic (resist simple frequency analysis)

Hill cipher
Uses matrix multiplication with key matrix (mod 26) encrypt block of letters

Key usage
Key matrix (eg: 2×2, 3×3 with number values)

complexity
High (requires matrix operations & modular arithmetic

security strength
very strong (harder to break, especially with larger matrices)

comparison Table

Feature	caesar cipher	monoalphabetic cipher	Vigenere cephure	Hill cipher
Key usage	single integer (shift value)	26 letter substition key	Keyword (repeating)	Key matrix. (eg: 2×2, 3×3)
complexity	very low	low to moderate	moderate	High
security Strength	weak	Moderate	stronger	very strong

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 20	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 25	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

